Practical Assignment 2

Problem Statement :

Write a program to store the first-year percentage of students in an array. Write function for sorting array of floating-point numbers in ascending order using-Selection Sort Bubble sort and display top five scores.

Input code:

```
#include <stdio.h>
#include <stdlib.h>
#define MAX_STUD 100

int main()
{
   float percentage[MAX_STUD];
   int num = 0, ch, i, j;
   float temp;

   do
   {
      printf("__Student percentage management system__\n");
      printf("1.Enter student percentage\n2.Selection sort\n3.Bubble sort\n4.Top 5\n5.Display all student percentage\n6.exit\n");
      printf("Enter your choice:");
      scanf("%d",&ch);
      switch(ch)
      {
         case 1:
            printf("Enter number of students (max %d): ", MAX_STUD);
            scanf("%d", &num);
            if(num > MAX_STUD || num <= 0)
```

```
    {
        printf("Insert a valid count!\n");
        num = 0;
    }
    else
    {
        printf("Enter percentages of students:\n");
        for(i = 0; i < num; i++)
        {
            printf("Percentage of student %d: ", i + 1);
            scanf("%f", &percentage[i]);
        }
    }
    break;

case 2:
    if(num == 0)
    {
        printf("No student data available! Please enter data first.\n");
        break;
    }
    for(i = 0; i < num - 1; i++)
    {
        int min = i;
        for(j = i + 1; j < num; j++)
        {
            if(percentage[j] < percentage[min])
            {
                min = j;
```

```
            }
        }
        if(min != i)
        {
            temp = percentage[i];
            percentage[i] = percentage[min];
            percentage[min] = temp;
        }
    }
    printf("\nPercentages sorted (Ascending) using Selection Sort:\n");
    for(i = 0; i < num; i++)
    {
        printf("%.2f%%\n", percentage[i]);
    }
    break;

case 3:
    if(num == 0)
    {
        printf("No student data available! Please enter data first.\n");
        break;
    }
    for(i = 0; i < num - 1; i++)
    {
        for(j = 0; j < num - 1 - i; j++)
        {
            if(percentage[j] > percentage[j + 1])
            {
                temp = percentage[j];
```

```
                percentage[j] = percentage[j + 1];
                percentage[j + 1] = temp;
            }
        }
    }
    printf("\nPercentages sorted (Ascending) using Bubble Sort:\n");
    for(i = 0; i < num; i++)
    {
        printf("%.2f%%\n", percentage[i]);
    }
    break;

case 4:
    if(num == 0)
    {
        printf("No student data available! Please enter data first.\n");
        break;
    }

    for(i = 0; i < num - 1; i++)
    {
        int min = i;
        for(j = i + 1; j < num; j++)
        {
            if(percentage[j] < percentage[min])
            {
                min = j;
            }
        }
```

```
            if(min != i)
            {
                temp = percentage[i];
                percentage[i] = percentage[min];
                percentage[min] = temp;
            }
        }

        printf("\nTop Percentages:\n");
        int count = 0;
        for(i = num - 1; i >= 0 && count < 5; i--)
        {
            printf("Rank %d: %.2f%%\n", count + 1, percentage[i]);
            count++;
        }
        break;

    case 5:
        if(num == 0)
        {
            printf("No student data available! Please enter data first.\n");
        }
        else
        {
            printf("\nAll Student Percentages:\n");
            for(i = 0; i < num; i++)
            {
                printf("Student %d: %.2f%%\n", i + 1, percentage[i]);
            }
```

```
            }
            break;

        case 6:
            printf("Exiting program...\n");
            break;

        default:
            printf("Invalid choice! Please enter a option between 1 and 6.\n");
            break;
      }
   } while(ch != 6);

   return 0;
}
```

Output:

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:1

Enter number of students (max 100): 5

Enter percentages of students:

Percentage of student 1: 98.6

Percentage of student 2: 56.7

Percentage of student 3: 86.9

Percentage of student 4: 82.1

Percentage of student 5: 70.0

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:2

Percentages sorted (Ascending) using Selection Sort:

56.70%

70.00%

82.10%

86.90%

98.60%

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:3

Percentages sorted (Ascending) using Bubble Sort:

56.70%

70.00%

82.10%

86.90%

98.60%

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:4

Top Percentages:

Rank 1: 98.60%

Rank 2: 86.90%

Rank 3: 82.10%

Rank 4: 70.00%

Rank 5: 56.70%

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:5

All Student Percentages:

Student 1: 56.70%

Student 2: 70.00%

Student 3: 82.10%

Student 4: 86.90%

Student 5: 98.60%

__Student percentage management system__

1.Enter student percentage

2.Selection sort

3.Bubble sort

4.Top 5

5.Display all student percentage

6.exit

Enter your choice:6

Exiting program...

Process returned 0 (0x0)   execution time : 51.723 s

Press any key to continue.